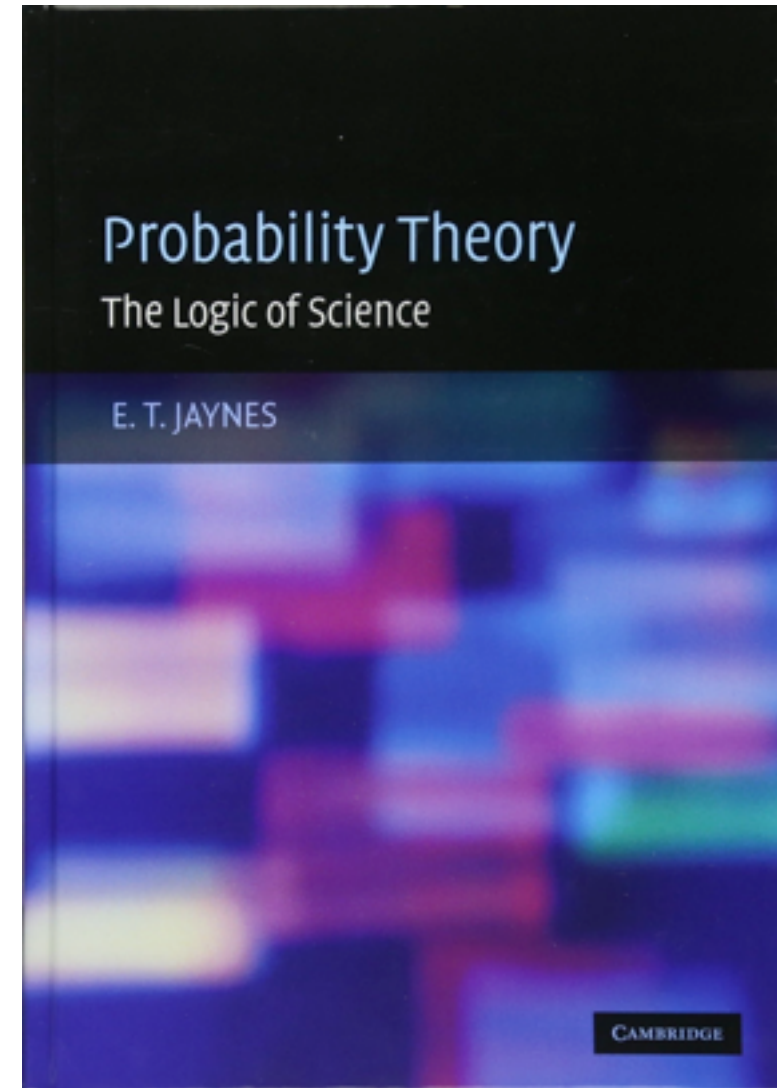




# Why normal?

- Ontological perspective
  - Processes which add fluctuations result in dampening
  - Damped fluctuations end up Gaussian
  - No information left, except mean and variance
  - Can't infer process from distribution!
- Epistemological perspective
  - Know only *mean* and *variance*
  - Then least surprising and most conservative (*maximum entropy*) distribution is Gaussian
  - Nature likes maximum entropy distributions



# Language for modeling

Give them definitions:

$$y_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \beta x_i$$

$$\beta \sim \text{Normal}(0, 10)$$

$$\sigma \sim \text{Exponential}(1)$$

$$x_i \sim \text{Normal}(0, 1)$$

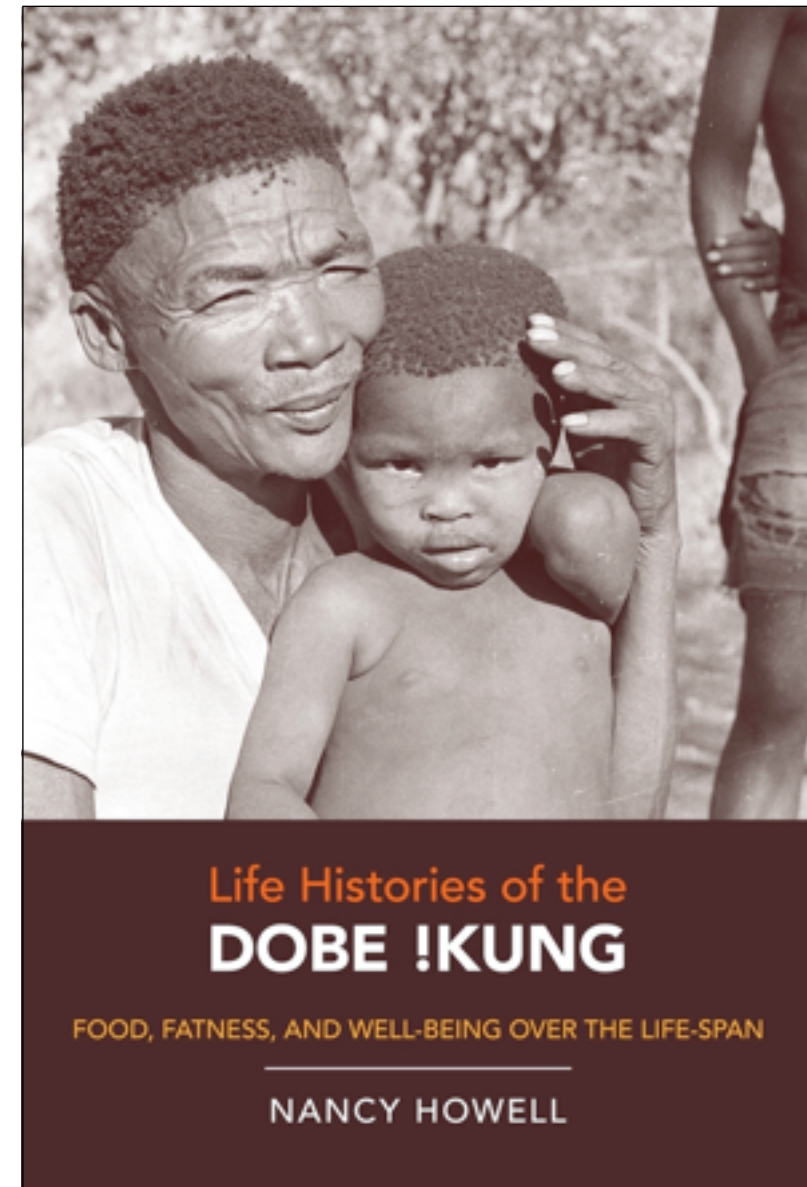
# Some data: Kalahari foragers

```
library(rethinking)  
data(Howell1)  
d <- Howell1
```

```
precis( d )
```

'data.frame': 544 obs. of 4 variables:

|        | mean   | sd    | 5.5%  | 94.5%  | histogram                                                                            |
|--------|--------|-------|-------|--------|--------------------------------------------------------------------------------------|
| height | 138.26 | 27.60 | 81.11 | 165.74 |  |
| weight | 35.61  | 14.72 | 9.36  | 54.50  |  |
| age    | 29.34  | 20.75 | 1.00  | 66.13  |  |
| male   | 0.47   | 0.50  | 0.00  | 1.00   |  |





# Adding a predictor variable

- Use a linear model of the mean,  $\mu$ :

$$h_i \sim \text{Normal}(\mu_i, \sigma) \quad [\text{likelihood}]$$

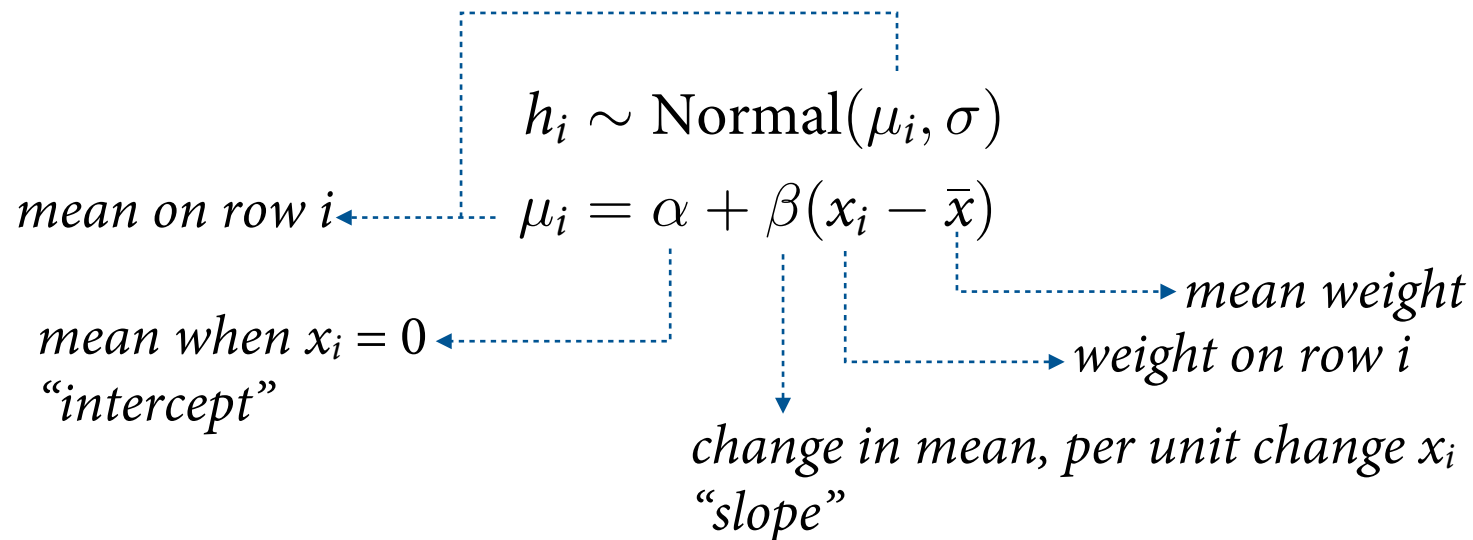
$$\mu_i = \alpha + \beta(x_i - \bar{x}) \quad [\text{linear model}]$$

$$\alpha \sim \text{Normal}(178, 20) \quad [\alpha \text{ prior}]$$

$$\beta \sim \text{Normal}(0, 10) \quad [\beta \text{ prior}]$$

$$\sigma \sim \text{Uniform}(0, 50) \quad [\sigma \text{ prior}]$$

# Adding a predictor variable



# Prior predictive distribution

```
set.seed(2971)
N <- 100                                # 100 lines
a <- rnorm( N , 178 , 20 )
b <- rnorm( N , 0 , 10 )
```

$$\alpha \sim \text{Normal}(178, 20)$$

$$\beta \sim \text{Normal}(0, 10)$$

R code  
4.38

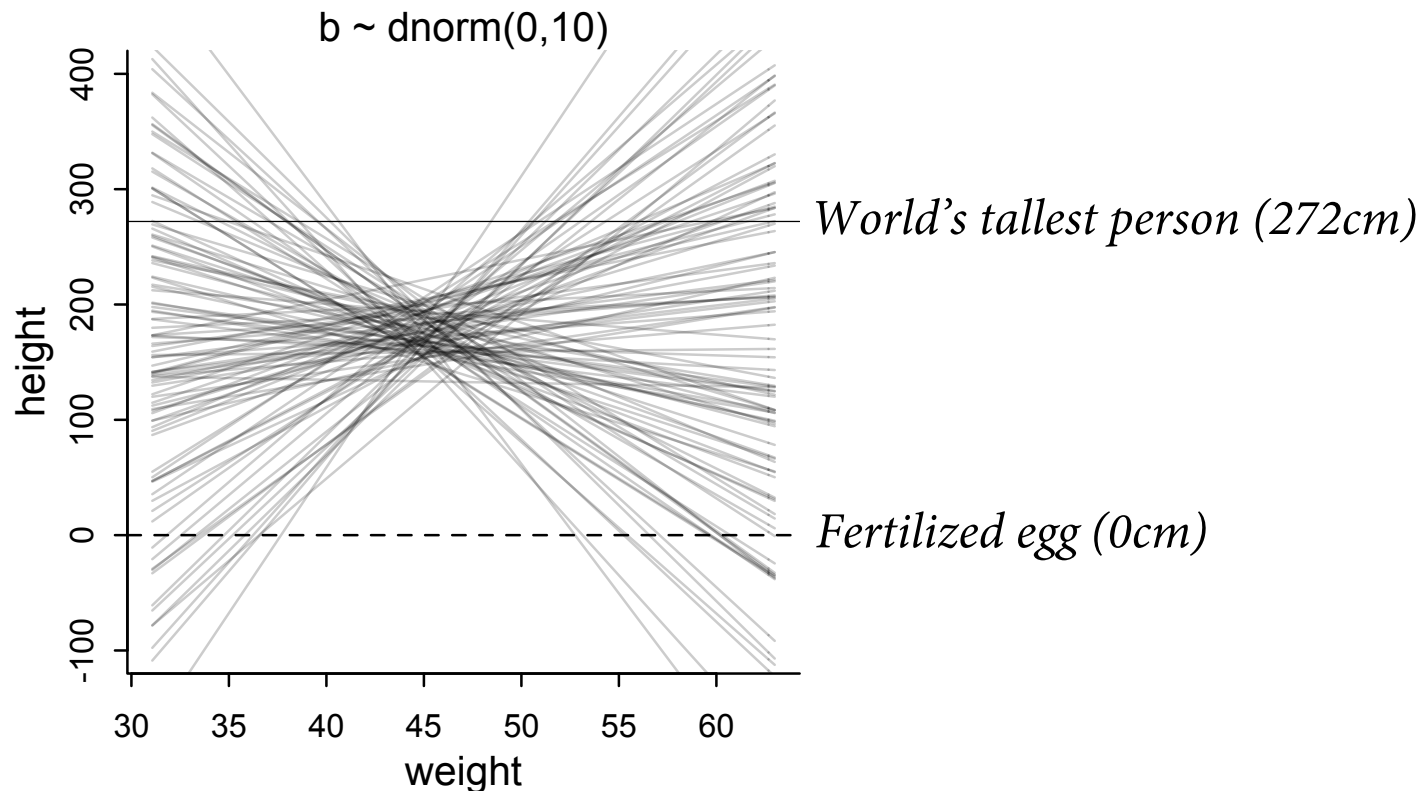


Figure 4.5

# Prior predictive distribution

R code  
4.41

```
set.seed(2971)
N <- 100                                # 100 lines
a <- rnorm( N , 178 , 20 )
b <- rlnorm( N , 0 , 1 )
```

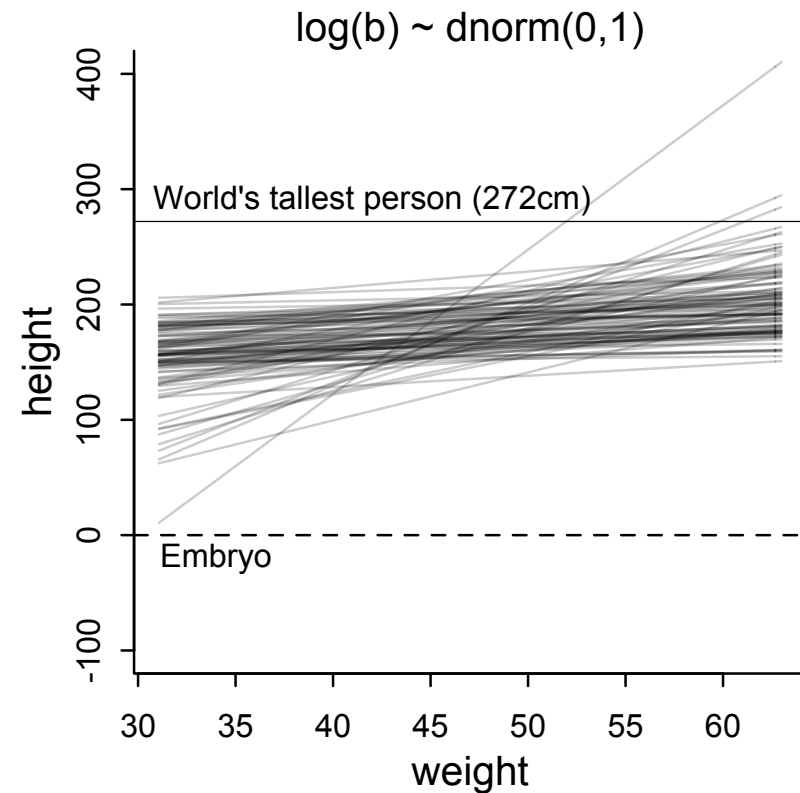
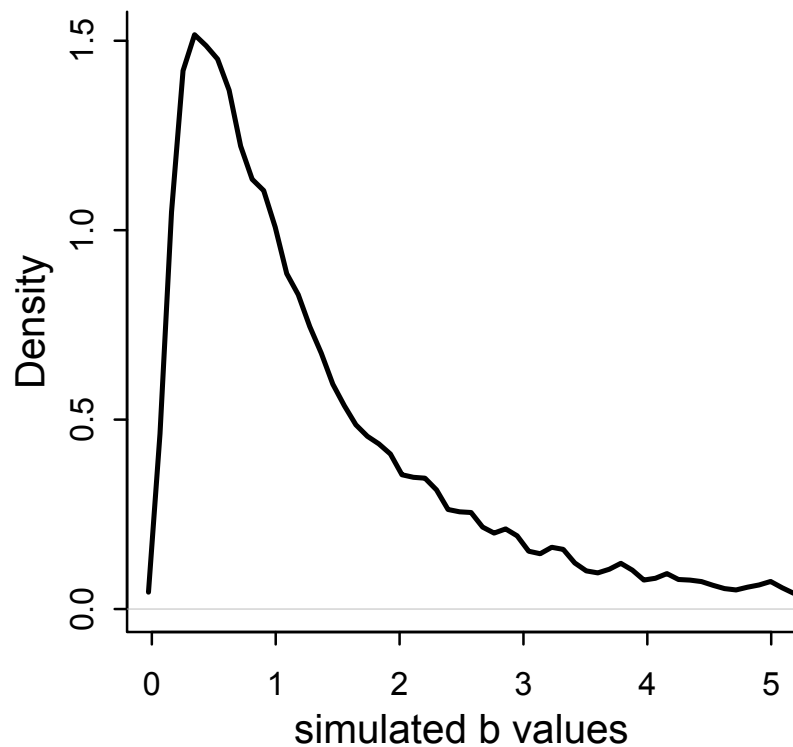


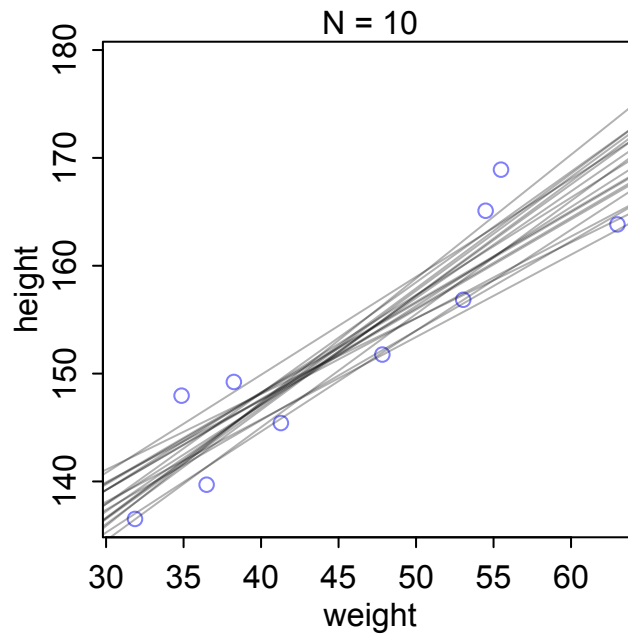
Figure 4.5



# Sampling from the posterior

- Want to get uncertainty onto that graph
- Again, sample from posterior
  1. Use mean and standard deviation to approximate posterior
  2. Sample from *multivariate normal* distribution of parameters
  3. Use samples to generate predictions that “integrate over” the uncertainty

# Posterior is full of lines



R code  
4.47

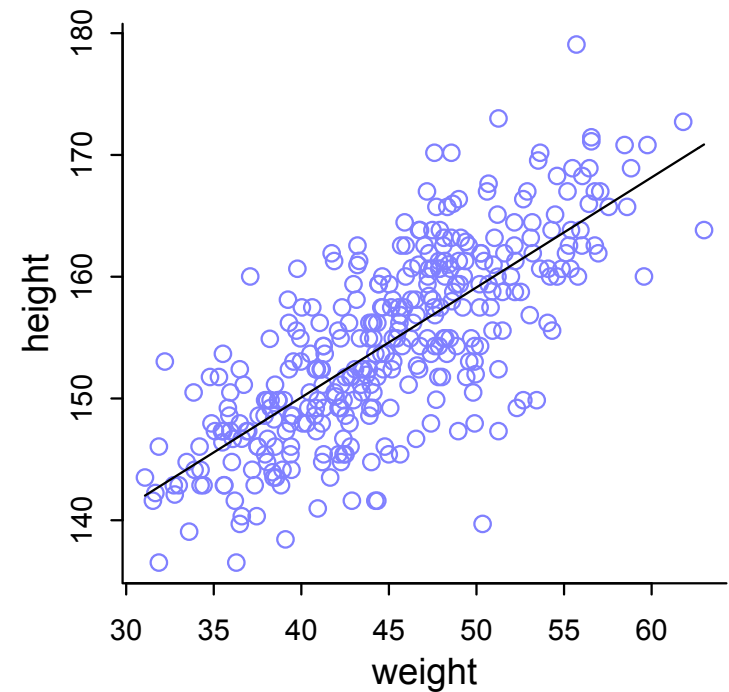
```
post <- extract.samples( m4.3 )  
post[1:5,]
```

|   | a        | b         | sigma    |
|---|----------|-----------|----------|
| 1 | 154.5505 | 0.9222372 | 5.188631 |
| 2 | 154.4965 | 0.9286227 | 5.278370 |
| 3 | 154.4794 | 0.9490329 | 4.937513 |
| 4 | 155.2289 | 0.9252048 | 4.869807 |
| 5 | 154.9545 | 0.8192535 | 5.063672 |

Figure 4.7

# Showing Uncertainty

- Want to get uncertainty onto that graph
- Again, sample from posterior
  1. Use mean and standard deviation to approximate posterior
  2. Sample from *multivariate normal* distribution of parameters
  3. Use samples to generate predictions that *integrate over* the uncertainty



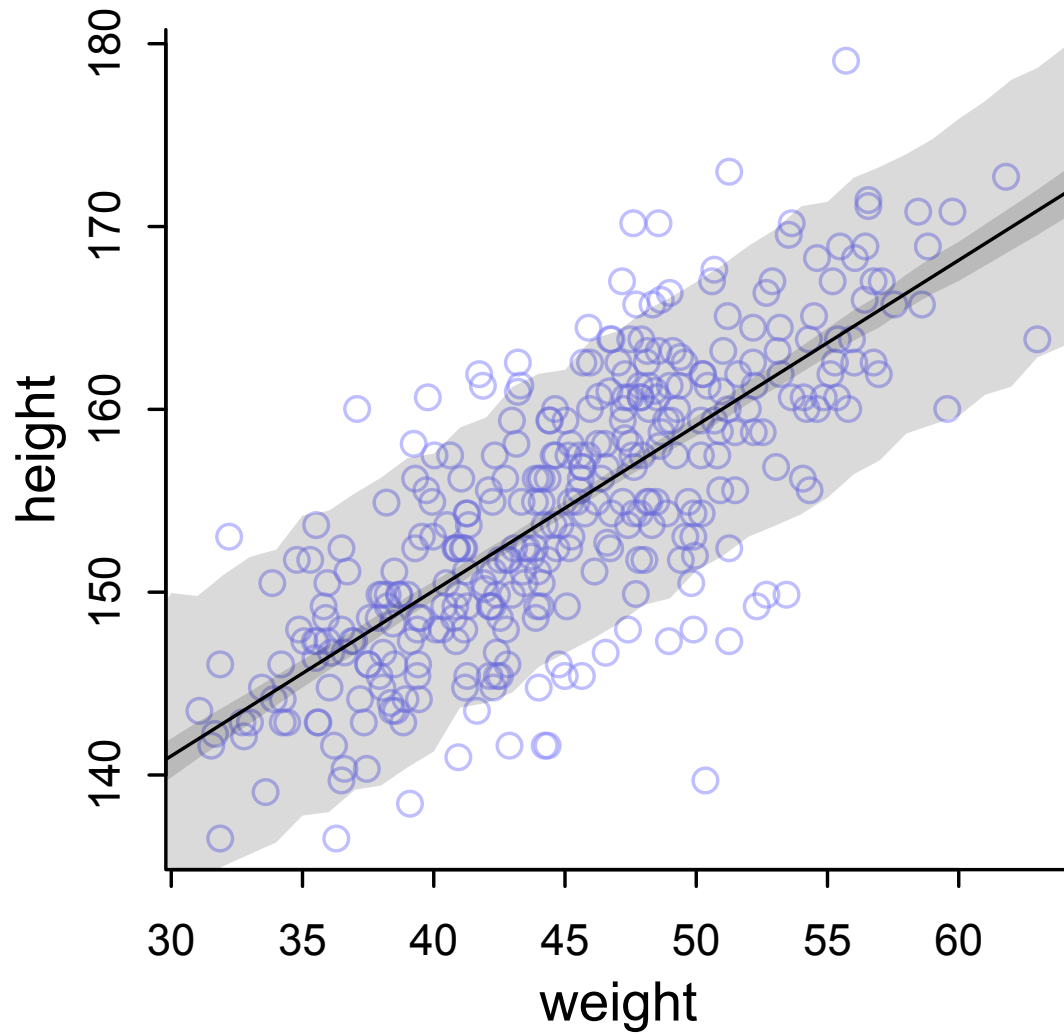
# How `link` works

- Sample from posterior
- Define series of predictor (weight) values
- For each predictor value
  - For each sample from posterior
    - Compute  $\mu$ :  $a + b \cdot (\text{weight} - \bar{x})$
- Summarize

R code  
4.58

```
post <- extract.samples(m4.3)
mu.link <- function(weight) post$a + post$b*( weight - xbar )
weight.seq <- seq( from=25 , to=70 , by=1 )
mu <- sapply( weight.seq , mu.link )
mu.mean <- apply( mu , 2 , mean )
mu.HPDI <- apply( mu , 2 , HPDI , prob=0.89 )
```

89% prediction interval



Nothing special about 95%

Try 50%, 80%, 99%

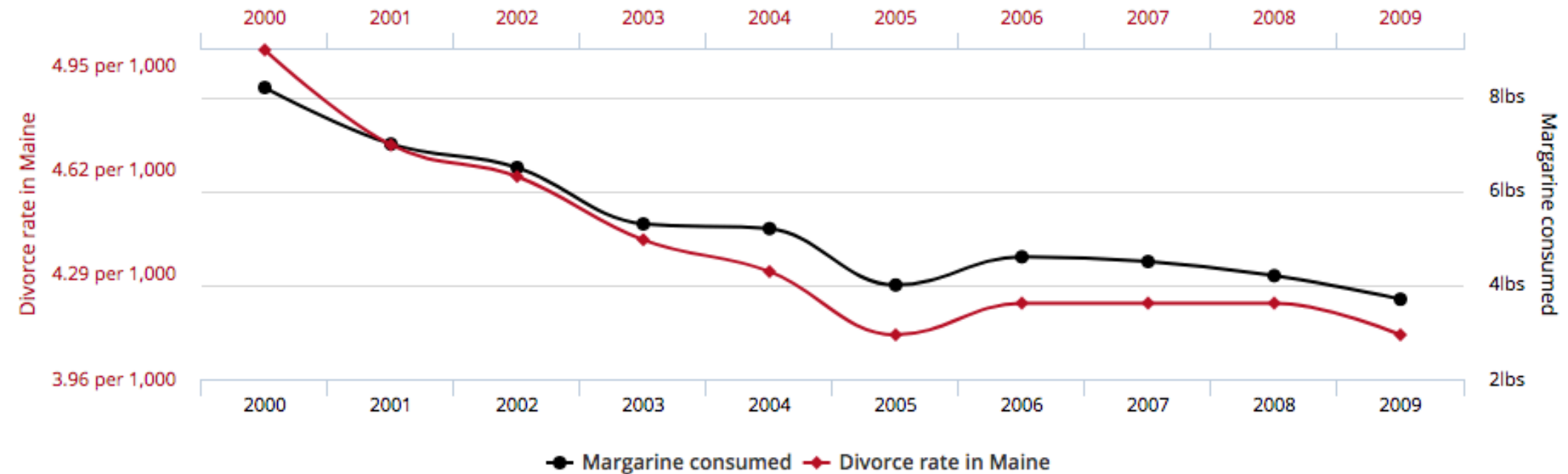
Interested in *shape*,  
not *boundaries*

Figure 4.10

# Correlation is commonplace

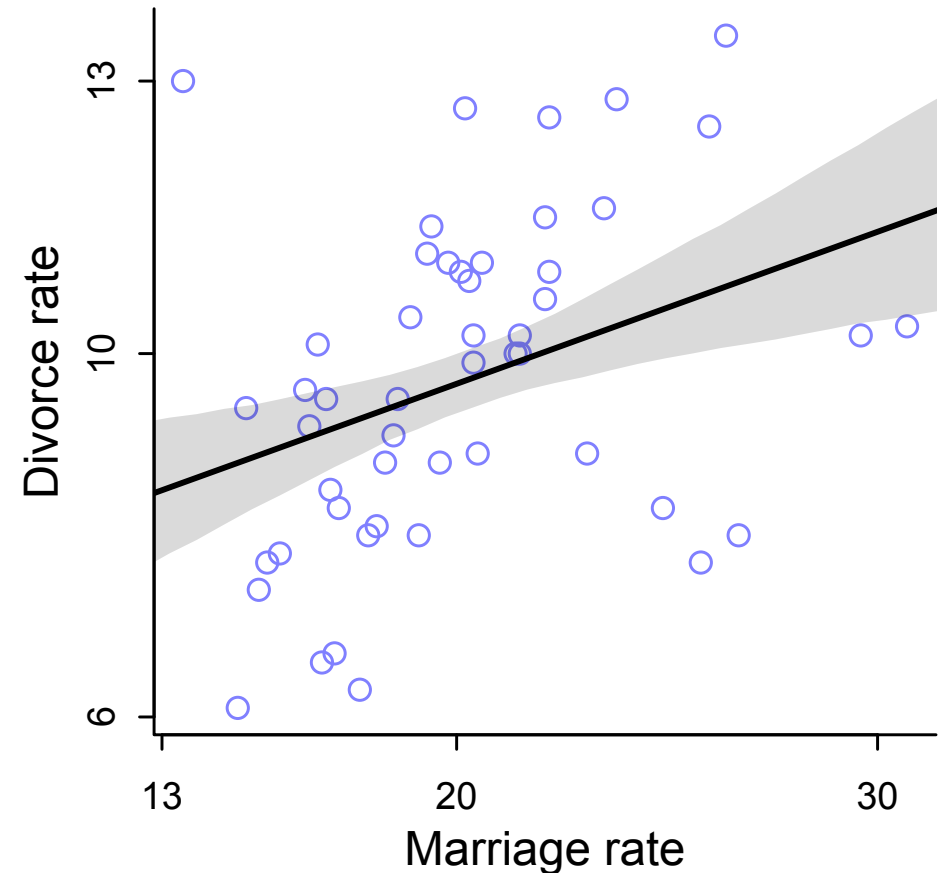
## Divorce rate in Maine correlates with Per capita consumption of margarine

Correlation: 99.26% ( $r=0.992558$ )



# Spurious association

- Correlation does not imply causation
- Causation does not imply correlation
- Causation implies conditional correlation
- Need more than just models
- Q: Does marriage cause divorce?



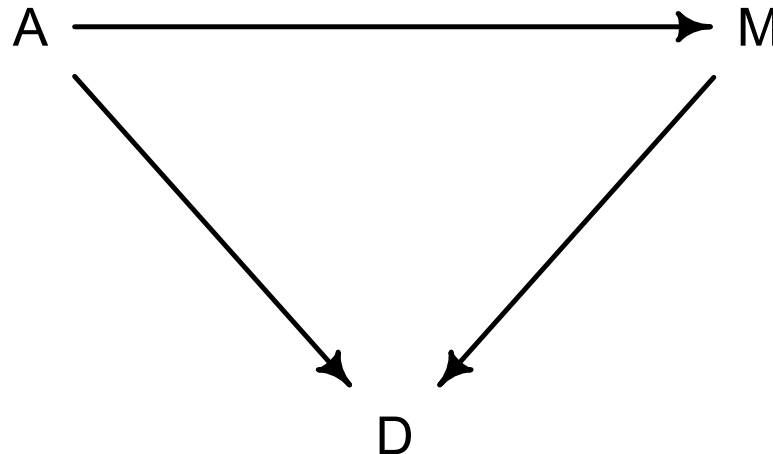


# Multiple causes of divorce

- Want to know: *what is value of a predictor, once we know the other predictors?*
  - What is value of knowing marriage rate, once we already know median age at marriage?
  - What is value of knowing median age marriage, once we know marriage rate?

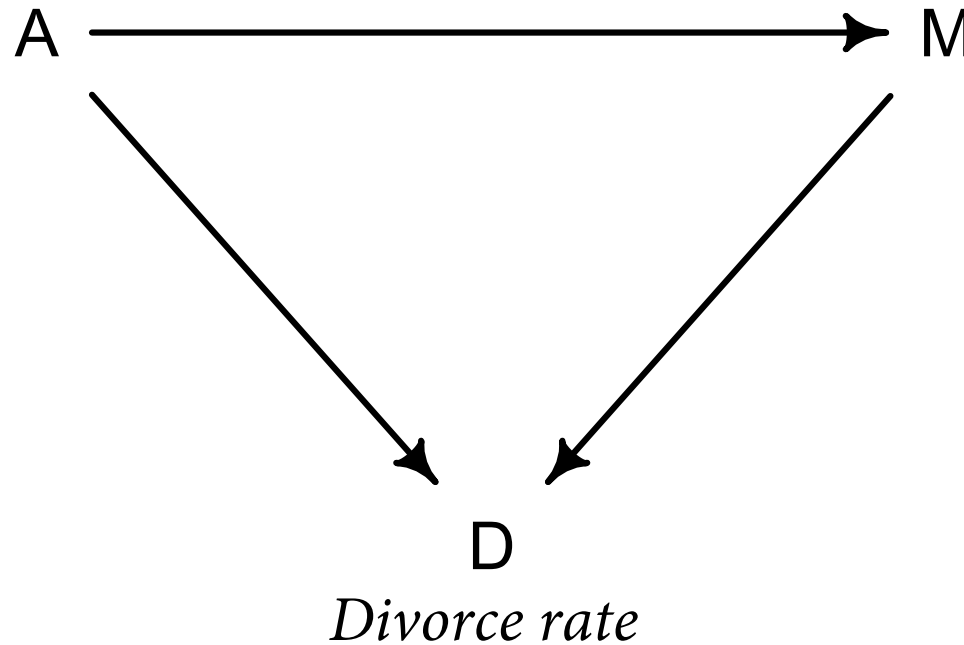
# They're good DAGs, Brent

- Directed Acyclic Graphs — tools for causal models
  - Directed: Arrows
  - Acyclic: Arrows don't make loops
  - Graphs: Nodes and edges
- Unlike statistical model, has causal implications



*Median age of marriage*

*Marriage rate*



Implications:

- (1) M is a function of A
- (2) D is a function of A and M
- (3) The total causal effect of A has two *paths*:
  - (a)  $A \rightarrow M \rightarrow D$
  - (b)  $A \rightarrow D$

*divorce rate*

$$D_i \sim \text{Normal}(\mu_i, \sigma)$$

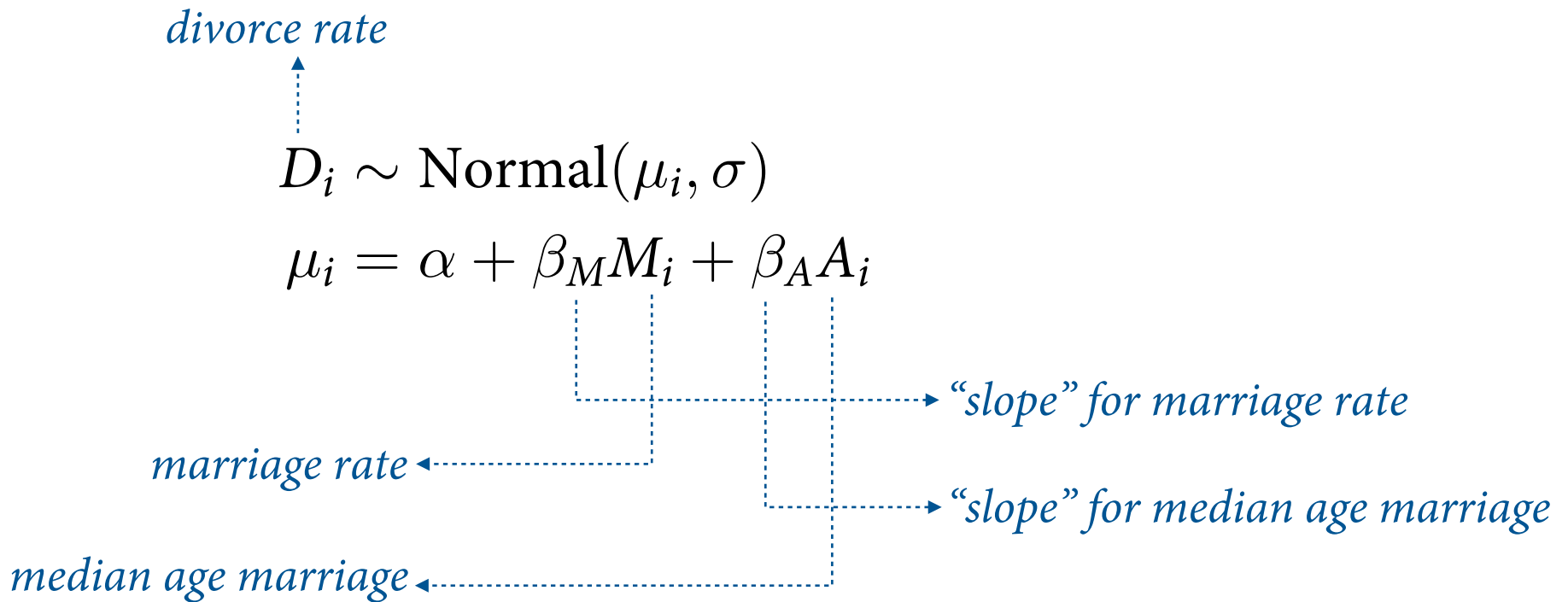
$$\mu_i = \alpha + \beta_M M_i + \beta_A A_i$$

*marriage rate*

*“slope” for marriage rate*

*“slope” for median age marriage*

*median age marriage*



$D_i \sim \text{Normal}(\mu_i, \sigma)$  [probability of data]

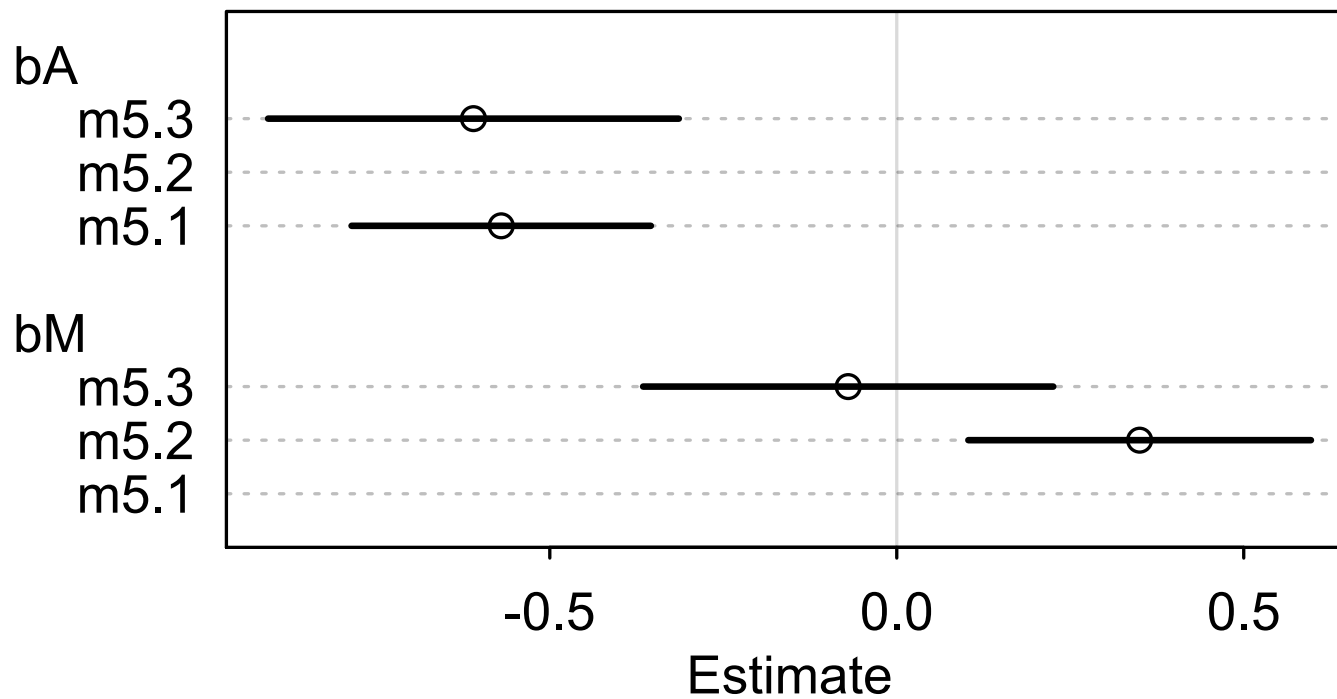
$\mu_i = \alpha + \beta_M M_i + \beta_A A_i$  [linear model]

$\alpha \sim \text{Normal}(0, 0.2)$  [prior for  $\alpha$ ]

$\beta_M \sim \text{Normal}(0, 0.5)$  [prior for  $\beta_M$ ]

$\beta_A \sim \text{Normal}(0, 0.5)$  [prior for  $\beta_A$ ]

$\sigma \sim \text{Exponential}(1)$  [prior for  $\sigma$ ]



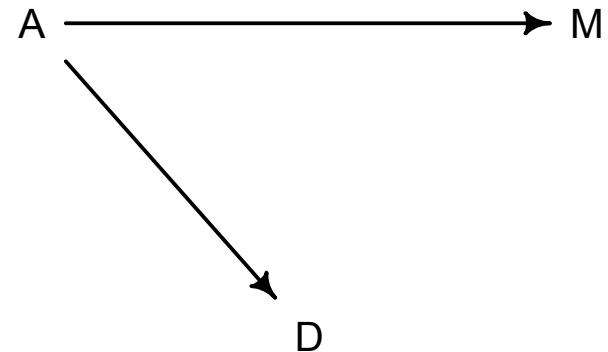
m5.1: age of marriage only  $D \sim A$

m5.2: marriage rate only  $D \sim M$

m5.3: multiple regression  $D \sim A + M$

# Multiple regression

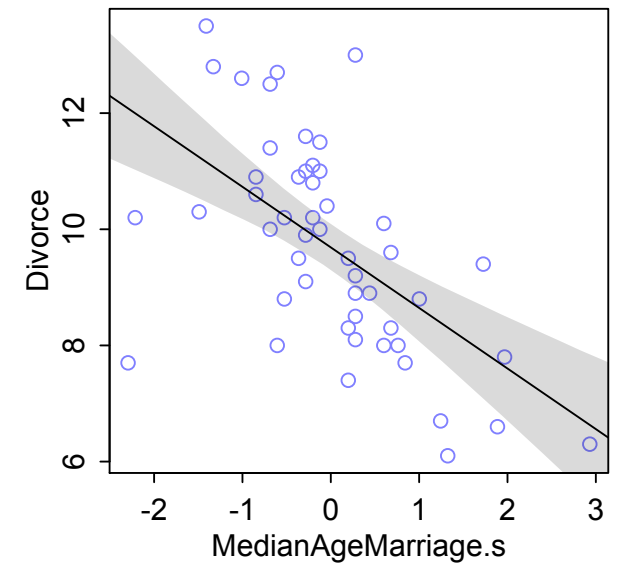
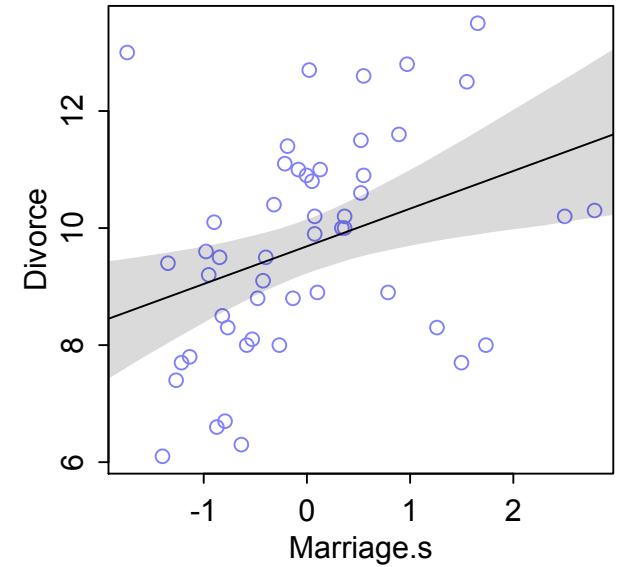
|       | mean  | sd   | 5.5%  | 94.5% |
|-------|-------|------|-------|-------|
| a     | 0.00  | 0.10 | -0.16 | 0.16  |
| bM    | -0.07 | 0.15 | -0.31 | 0.18  |
| bA    | -0.61 | 0.15 | -0.85 | -0.37 |
| sigma | 0.79  | 0.08 | 0.66  | 0.91  |



- Once we know median age marriage, little additional value in knowing marriage rate.
- Once we know marriage rate, still value in knowing median age marriage.
- If we *don't know* median age marriage, still useful to know marriage rate.

# Posterior predictions

- Lots of plotting options now
  1. Predictor residual plots
  2. Counterfactual plots
  3. Posterior prediction plots





# Masked association

- Sometimes association between outcome and predictor masked by another variable
- Need both variables to see influence of either
- Tends to arise when
  - Another predictor associated with outcome *in opposite direction*
  - Both predictors associated with one another
- Noise in predictors can also mask association (*residual confounding*)





# Index variable

R code  
5.36

```
d$sex <- ifelse( d$male==1 , 2 , 1 )  
str( d$sex )
```

```
num [1:544] 2 1 1 2 1 2 1 2 1 2 ...
```

$$h_i \sim \text{Normal}(\mu_i, \sigma)$$

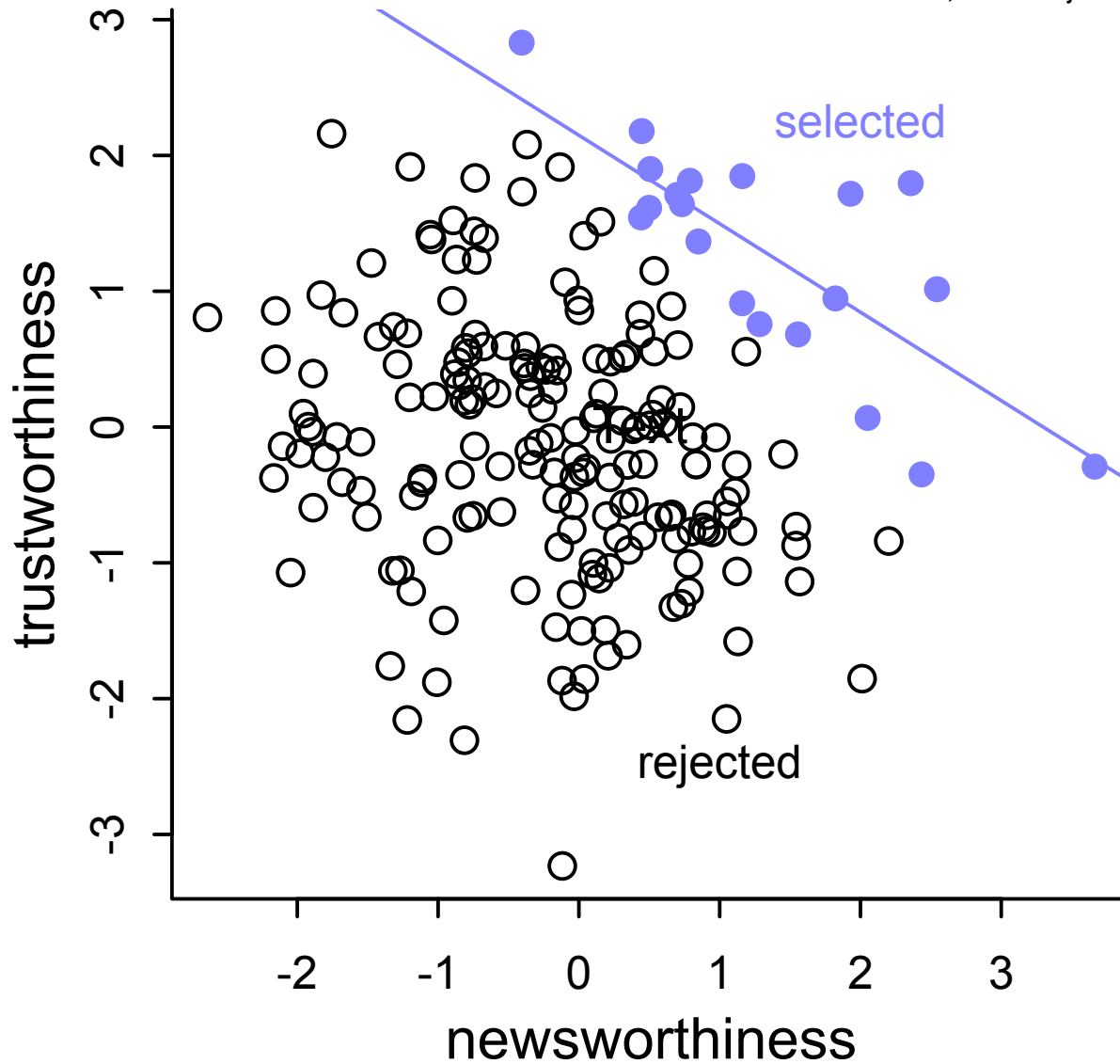
$$\mu_i = \alpha_{\text{SEX}[i]}$$

$$\alpha_j \sim \text{Normal}(178, 20) \quad , \text{ for } j = 1..2$$

$$\sigma \sim \text{Uniform}(0, 50)$$

# Selection-distortion effect

when a sample is selected on a combination of 2 (or more) variables, the relationship between those 2 variables is different after selection than it was before, and not just because of restriction of range



# Why not just add everything?

- Could just add all available predictors to model
  - “We controlled for...”
- Almost always a bad idea
  - Adding variables *creates* confounds
  - Residual confounding
  - Overfitting



## *The Fork*



Open unless you  
condition on Z

## *The Pipe*



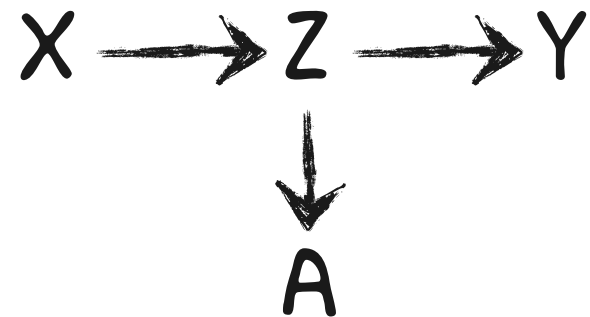
Open unless you  
condition on Z

## *The Collider*



Closed until you  
condition on Z

## *The Descendant*

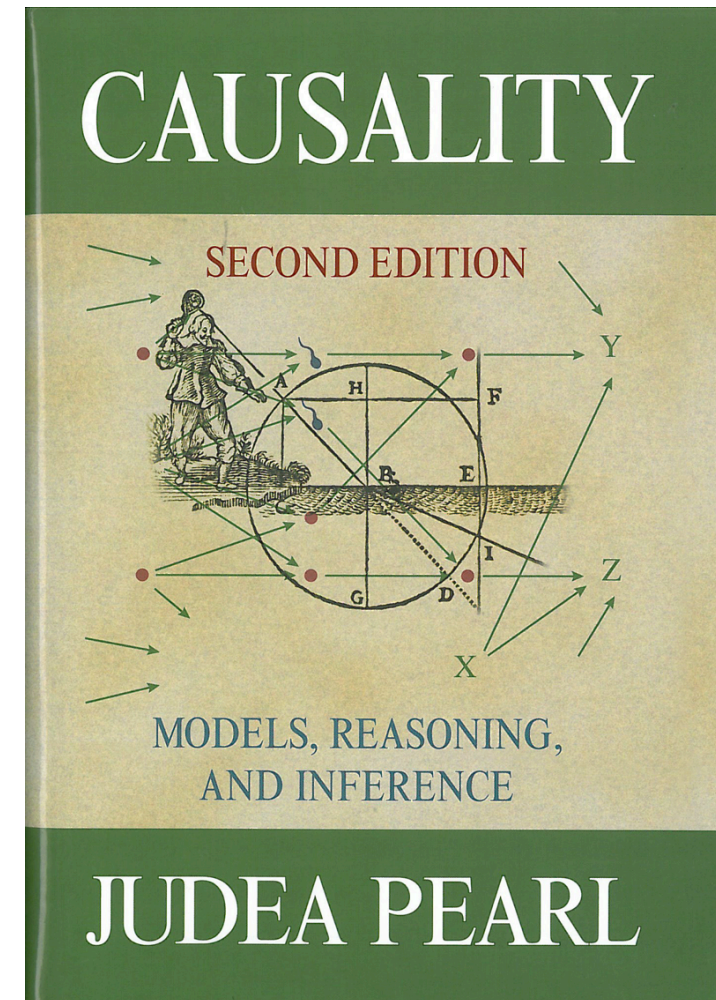
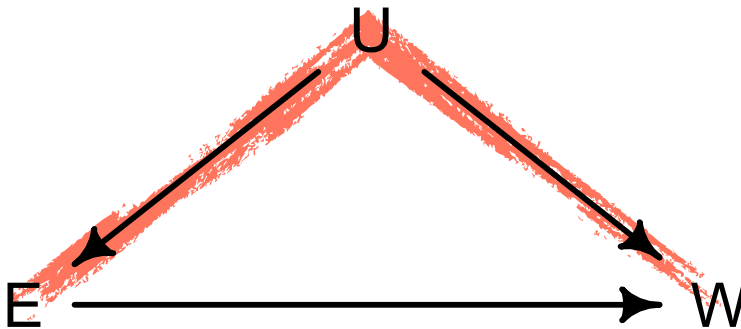


Conditioning on A is  
like conditioning on Z



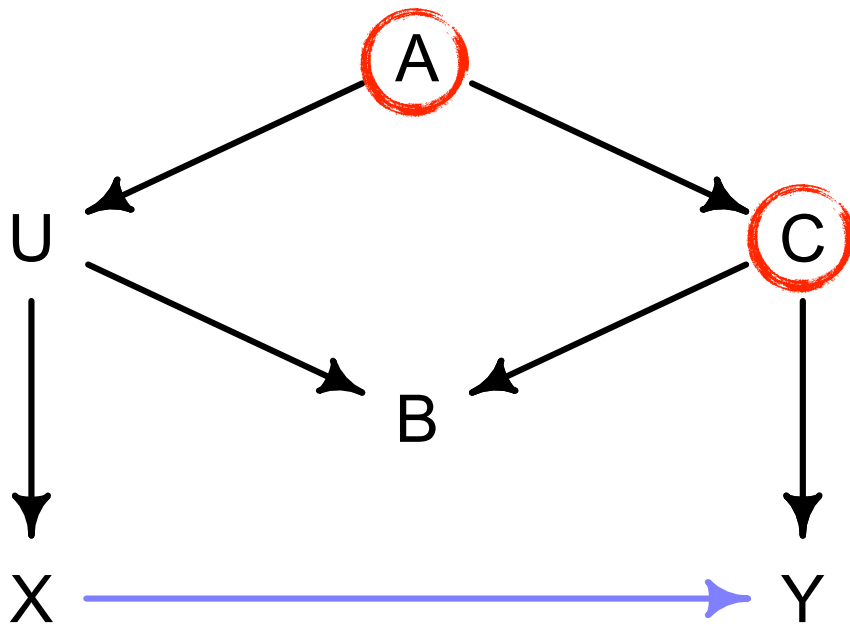
# Shutting the back door

- What ties these examples together:
- The **back-door criterion**: Confounding caused by existence of open back door paths from X to Y
- If you know your elements, you know how to open/close each of them



# Something more interesting

- Which variables, if any, should you condition on to infer  $X \rightarrow Y$ ?
- Condition on  $A$  or  $C$ . Do not condition on  $B$ .



(1)  $X \leftarrow U \leftarrow A \rightarrow C \rightarrow Y$   
This path is open.

(2)  $X \leftarrow U \rightarrow B \leftarrow C \rightarrow Y$   
This path is closed.



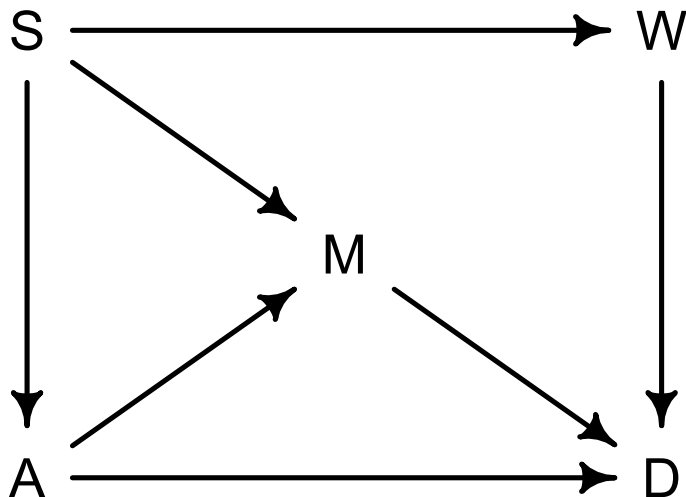
# Implied conditional independence

- Given DAG, can test some implications

```
impliedConditionalIndependencies( dag_6.2 )
```

R code  
6.36

```
A _||_ W | S
D _||_ S | A, M, W
M _||_ W | S
```



- (1) A and W independent, conditioning on S
- (2) D and S independent, conditioning on A, M, & W
- (3) M and W independent, conditioning on S

# Causal inference hard but possible

- Demonstrate capable of inferring cause
- Experiments not required!
- Experiments not always practical & ethical
  - Disease, evolution, development, dynamics of popular music, global climate, war
- Experiments must choose an intervention
  - Interventions influence many variables at once
  - Experimentally manipulate obesity?

